

7. Integration Techniques

7.4. Partial Fraction Decomposition

7.4.3. Integrals Needing Partial Fraction Decomposition (Continued)

7.4b Learning Objectives

- I can use partial fraction decomposition to help me solve integrals.

Example 7.4.1

Compute the following antiderivative

$$\int \frac{3e^{2x} - 5e^{3x}}{(3e^x - 1)(e^x - 1)^2} dx$$

Note

If you feel like you need to review exponent rules do that, note

$$e^{3x} = e^{2x} e^x$$

for example.

Solution

First, it looks like e^x is everywhere, so we will do a quick u -substitution with $u = e^x$. Then $du = e^x dx$

$$\begin{aligned} \int \frac{3e^{2x} - 5e^{3x}}{(3e^x - 1)(e^x - 1)^2} dx &= \int \frac{3e^x - 5e^{2x}}{(3e^x - 1)(e^x - 1)^2} e^x dx \\ &= \int \frac{3u - 5u^2}{(3u - 1)(u - 1)^2} du \end{aligned}$$

Now, this integral needs a partial fraction decomposition

Example 7.4.2

Compute the following antiderivative

$$\int \frac{x^4 + 2x^2 + 1}{x - 1} dx$$

7.4b Section Summary:

- We solved a lot of integrals using partial fraction decomposition.